

6	(a)	$\Phi = NBA = 400 \times 5 \times 10^{-5} \times \sin 60^\circ \times 25 \times 10^{-4}$ $= 4.33 \times 10^{-5} \text{ Wb}$		C1 A1
	(b)	(i)	$\Delta\Phi = \Phi_f - \Phi_i = [4.33 - (-4.33)] \times 10^{-5} = 8.66 \times 10^{-5} \text{ Wb}$	A1
		(ii)	Induced emf $E = \left \frac{d\Phi}{dt} \right = \frac{\Delta\Phi}{\Delta t}$ $= \frac{8.66 \times 10^{-5}}{2.5}$ $= 3.46 \times 10^{-5} \text{ V}$	C1 A1
	(c)	Clockwise direction from the top view.		A1
	(d)	According to Lenz's Law, the induced current flows in a direction so as to oppose the change that produces it. <u>This will result in a magnetic force that opposes the rotation of the coil.</u>		B1
		Mechanical energy / kinetic energy is converted to electrical energy.		B1